Problem 1: Given the vectors $\mathbf{v} = (3, 2)$ and $\mathbf{w} = (-1, 4)$, find scalars a and b such that $a\mathbf{v} + b\mathbf{w} = (7, -6)$.

Solution

Set up the equation:

$$a(3, 2) + b(-1, 4) = (7, -6).$$

This yields:

(i) $3a - b = 7$,

(ii) $2a + 4b = -6$.

Multiply (i) by 4:

$$12a - 4b = 28.$$

Add to (ii):

$$(12a - 4b) + (2a + 4b) = 28 + (-6)$$

$$14a = 22$$

$$a = \frac{22}{14} = \frac{11}{7}.$$

Substitute into (i):

$$3\left(\frac{11}{7}\right) - b = 7$$

$$b = \frac{33}{7} - 7 = -\frac{16}{7}.$$

Thus:

$$\boxed{a = \frac{11}{7},\ b = -\frac{16}{7}}$$

Problem 2: Determine whether the vectors $\mathbf{v} = (1, 2, 3), \mathbf{w} = (2, 3, 4)$, and $\mathbf{u} = (3, 4, 5)$ spans $\mathbb{R}^3$.

Solution

Form the matrix A with the vectors as columns:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1\begin{vmatrix} 3 & 4 \\ 4 & 5 \end{vmatrix} - 2\begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} + 3\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix}$$

$$= 1(15 - 16) - 2(10 - 12) + 3(8 - 9)$$

$$= -1 + 4 - 3 = 0.$$

Since $\det(A) = 0$, the vectors are linearly dependent and do not span $\mathbb{R}^3$.

$$\boxed{\text{No, the vectors do not span } \mathbb{R}^3}$$

Problem 3: Given the vectors $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (-1, 3)$, find scalars a and b such that $a\mathbf{v} + b\mathbf{w} = (5, 0)$.

Solution

Set up the equation:

$$a(2, 1) + b(-1, 3) = (5, 0).$$

This yields:

(i) $2a - b = 5$,

(ii) $a + 3b = 0$.

From (ii): $a = -3b$. Substitute into (i):

$$2(-3b) - b = 5$$

$$-7b = 5$$

$$b = -\frac{5}{7}.$$

Then:

$$a = -3\left(-\frac{5}{7}\right) = \frac{15}{7}.$$

Thus:

$$\boxed{a = \frac{15}{7},\ b = -\frac{5}{7}}$$

Problem 4: Determine whether the vector $\mathbf{u} = (4, -2)$ can be expressed as a linear combination of $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, -1)$.

Solution

Solve $a(1, 2) + b(3, -1) = (4, -2)$. This yields:

(i) $a + 3b = 4$,

(ii) $2a - b = -2$.

Multiply (ii) by 3:

$$6a - 3b = -6.$$

Add to (i):

$$7a = -2$$
$$a = -\frac{2}{7}.$$

Substitute into (i):

$$-\frac{2}{7} + 3b = 4$$
$$3b = \frac{30}{7}$$
$$b = \frac{10}{7}.$$

Since scalars exist, $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

$$\boxed{\text{Yes}}$$

Problem 5: Determine whether $\mathbf{v} = (1, 0, 1)$, $\mathbf{w} = (0, 1, 1)$, and $\mathbf{u} = (1, 1, 0)$ span $\mathbb{R}^3$.

Solution

Form the matrix A with the vectors as columns:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= 1(0 - 1) + 1(0 - 1) = -1 - 1 = -2.$$

Since $\det(A) \neq 0$, the vectors are linearly independent and span $\mathbb{R}^3$.

$$\boxed{\text{Yes, the vectors span } \mathbb{R}^3}$$

Problem 6: Determine whether $\mathbf{u} = (6, -3)$ can be expressed as a linear combination of $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (2, -1)$.

Solution

Solve $a(1, 2) + b(2, -1) = (6, -3)$. This yields:

(i) $a + 2b = 6$,

(ii) $2a - b = -3$.

Multiply (ii) by 2:

$$4a - 2b = -6.$$

Add to (i):

$$5a = 0$$

$$a = 0.$$

Substitute into (i):

$$2b = 6$$

$$b = 3.$$

Since scalars exist, $\mathbf{u}$ is a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

$$\boxed{\text{Yes}}$$

Problem 7: Determine whether $\mathbf{u} = (5, 1)$ can be expressed as a linear combination of $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (4, 2)$.

Solution

Solve $a(2, 1) + b(4, 2) = (5, 1)$. This yields:

(i) $2a + 4b = 5$,

(ii) $a + 2b = 1$.

Multiply (ii) by 2:

$$2a + 4b = 2.$$

Subtract from (i):

$$(2a + 4b) - (2a + 4b) = 5 - 2$$

$$0 = 3.$$

The contradiction implies no solution exists. Thus, $\mathbf{u}$ is not a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

$$\boxed{\text{No}}$$

Problem 8: Determine whether $\mathbf{v} = (1, 1, 0)$, $\mathbf{w} = (0, 1, 1)$, and $\mathbf{u} = (1, 0, 1)$ span $\mathbb{R}^3$.

Solution

Form the matrix A with the vectors as columns:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Compute the determinant:

$$\det(A) = 1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= 1(1 - 0) + 1(1 - 0) = 1 + 1 = 2.$$

Since $\det(A) \neq 0$, the vectors are linearly independent and span $\mathbb{R}^3$.

$$\boxed{\text{Yes, the vectors span } \mathbb{R}^3}$$